# Full Stack IoT Development: CIS4930

## Assignment: ESP32 Audio Classification with Edge Impulse

In this assignment, we'll train a machine learning model to recognize a custom phrase ( "Hello, [Your Name"). You'll then deploy it to your ESP32 and use an INMP441 microphone to recognize the phrase in real time.

Materials:

1. ESP32
2. INMP441 I2S Microphone (Recieve in class)
3. Arduino IDE

## Part 1: Edge Impulse

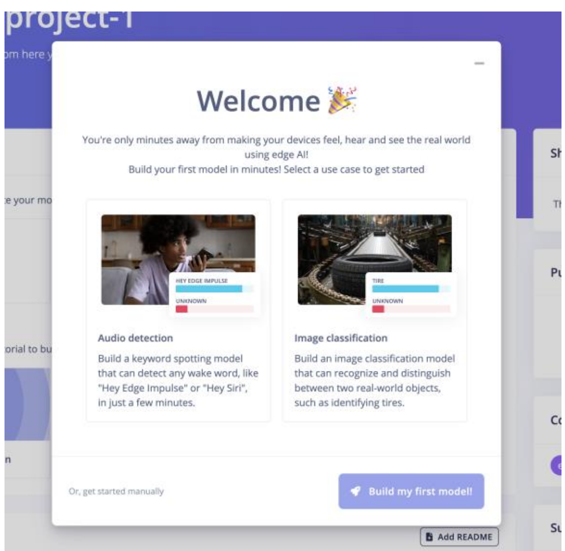
You'll need to create an Edge Impulse Account by doing the following:

- Go to https://edgeimpuulse.com
- Click "Sign up" and create a new free account
- After signing up, you'll be prompted with a message to create and build your first ML Model.

## Part 2: Follow the guided steps

You'll be prompted to walk through steps that include the following processes:

- Data Collection
- Impulse Design
- Model Training
- Model Testing



## Part 3: Connecting your ESP32

Connect your INMP441 Microphone using the either a breadboard jumper cables:

- VDD > 3.3V
- GND > GND
- SD > GPIO33
- SCK > GPIO 14
- WS > GPIO32
- L/R > 3.3V

A modification will be required to use pin 14 as your SCK.

## Part 3: Deploying to your ESP32

Use the following instructions to download the library necessary for installing on your Arduino:

1. Navigate to "deployment" in the left sidebar
2. Under deployment target, select Arduino Library
3. Click build and download the .zip file somewhere you can access it (e.g Downloads or Desktop)

Next, you'll need to properly setup Arduino IDE with your library using the following instructions:

1. Navigate to Sketch > Include Library > Add. ZIP library
2. Navigate to the .zip file you downloaded from Edge Impulse in the previous step
3. Arduino IDE will install the library, and you'll be ready to upload the code inside the zip file.

Finally, to upload your code follow the steps below:

1. Navigate to File > Examples
2. Locate your library name (it'll be the same as the ZIP file downloaded in the previous step)
3. Navigate to esp32 > esp32_microphone
4. Open the Sketch
5. In the sketch, modify the **i2s_pin_config_t** and set bck_io_num to pin 14 instead of 26.
6. Next, upload the code similar to how you'd upload any of your other ESP code.

## Part 6: Adding extra functionality

Now that you've got your code and microphone successfully working, use the following instructions to add some extra functionality:

1. Use an LED from your kit and connect it to an available GPIO port using a 220Ω resistor.
2. Modify your code such that if the wake word is detected, the LED turns on for 2 seconds
3. Hint: Modify your code near the prediction output to compare the result.classification value against 0.5 and toggle the LED.

## Part 5: Submission Instructions

For your submission upload a video as unlisted to YouTube where you complete the following:

- Open the Serial Monitor and ensure your baud rate is set to 115200
- If you've completed the steps correctly you should see an output depicting the recording
- Test your wake phrase 6-12 inches away from the microphone and observe the output
- Your prediction for the "hey_name" category should be greater than 0.5